

THE UNIVERSITY OF AZAD JAMMU & KASHMIR
MUZAFFARABAD

DATABASE MANAGEMENT SYSTEM

Lab 2 — Point of Sale System

Point of Sale (POS) Database

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Roll No	2024-SE-11
Semester	04
Course Title	Database Management System (CS-2204)
Submitted To	Engr. Awais Rathore
Lab No.	02

1. Objective

- Design a complete Point of Sale database covering security/access control, inventory, and sales.
- Implement Role-Based Access Control (RBAC) for Admins, Salesmen, and Customers.
- Track products, categories, and stock adjustments through an inventory audit log.
- Record orders, order line items, and time-based discount rules.
- Write analytical reporting queries to extract business insights from the data.

2. Database Structure

The database (Point_of_Sale) is organized into three functional groups of tables, for a total of 10 tables.

2.1 Security & Auth

Table	Purpose
roles	Defines system roles (Admin, Salesman, Customer)
permissions	Defines individual system permissions
role_permissions	Junction table linking roles to permissions
users	System users, linked to a role

2.2 Inventory

Table	Purpose
categories	Product categories
products	Product details, price, and category
inventory_logs	Audit trail of every stock adjustment, by user

2.3 Sales & Billing

Table	Purpose
orders	Order header — customer and order date
order_items	Line items linking products to orders, with quantity
discount_rules	Time-sensitive promotional discounts per product

3. Key Tasks / Concepts Implemented

- Creating all 10 tables with appropriate primary keys, foreign keys, and relationships.
- Inserting realistic sample data across roles, users, products, and orders.
- Query 1-3: Revenue tracking — total spent per customer, total revenue per category.
- Query 4-6: Inventory insights — best-selling products by quantity, full stock audit trail.
- Query 7-8: Sales operations — comparing discounted vs. non-discounted order value.
- Query 9-10: Role/permission reporting — which users hold which permissions.

4. Implementation (phpMyAdmin)

The database and all 10 reporting queries were implemented and executed using phpMyAdmin under XAMPP.

Table	Action	Rows	Type	Collation	Size	Overhead
☐ categories	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ discount_rules	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ inventory_logs	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ orders	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ order_items	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ permissions	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ products	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ roles	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ role_permissions	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ users	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
10 tables	Sum	0	InnoDB	utf8mb4_general_ci	368.0 KiB	0 B

Structure showing all 10 created tables

Server: 127.0.0.1 » Database: point_of_sale » Table: orders

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	id	user_id	discount_id	total_amount	created_at
☐ Edit Copy Delete	1	6	10	1530.00	2025-04-01 10:15:00
☐ Edit Copy Delete	2	7	NULL	850.00	2025-04-02 11:30:00
☐ Edit Copy Delete	3	8	5	2812.50	2025-04-03 14:00:00
☐ Edit Copy Delete	4	9	3	900.00	2025-04-04 09:45:00
☐ Edit Copy Delete	5	10	NULL	1200.00	2025-04-05 16:20:00
☐ Edit Copy Delete	6	6	6	585.00	2025-04-06 12:00:00
☐ Edit Copy Delete	7	7	8	3080.00	2025-04-07 13:10:00
☐ Edit Copy Delete	8	8	NULL	340.00	2025-04-08 15:30:00
☐ Edit Copy Delete	9	9	2	3972.50	2025-04-09 10:00:00
☐ Edit Copy Delete	10	10	7	1011.60	2025-04-10 17:45:00

☐ Check all | With selected: Edit Copy Delete Export

Browse table showing sample data (of Orders table)

Server: 127.0.0.1 » Database: point_of_sale » Table: orders

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

id	product_name	category_name	price
9	Whole Wheat Bread	Bakery	110.00
5	Mineral Water 1.5L	Beverages	60.00
10	Dishwashing Liquid 500ml	Cleaning Supplies	160.00
3	Mens Polo Shirt	Clothing	950.00
8	Full Cream Milk 1L	Dairy Products	180.00
2	USB-C Hub	Electronics	1200.00
1	Wireless Mouse	Electronics	850.00
7	Running Shoes	Footwear	3500.00
4	Basmati Rice 5kg	Groceries	650.00
6	Ballpoint Pen (12-pack)	Stationery	120.00

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Output of a reporting query that “Maps every product to its respective category”

5. Conclusion

This lab implemented a realistic, RBAC-secured Point of Sale database and demonstrated how relational joins and aggregation can turn raw transactional data into meaningful business reports.